

TOBY IO

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Summary

Applied AI engineer with 5+ years shipping full-stack systems end to end, from 20+ platforms and AI pipelines across manufacturing, hospitality, and government, to LLM copilots (RAG, agents, voice) on live factory floors.

Technical Skills

Languages: Python, TypeScript, SQL, JavaScript

Frameworks: React, Next.js, Node.js, Medusa.js, TanStack, Prisma, Flask, FastAPI

AI/ML: PyTorch, LangChain, PydanticAI, RAG, MCP, LoRA, LiveKit, ElevenLabs, Vercel AI SDK, Stable Diffusion

Infrastructure: AWS, Azure, Docker, PostgreSQL, Redis, Kafka, Cloudflare, Vercel

Work Experience

Taktora Inc. | *Website*

Nov 2024 – May 2026

Co-Founder & CTO

Los Angeles, CA

- Co-founded Taktora, an AI copilot ('Cursor for factory scheduling') that adapts production plans to real-time floor conditions. Deployed the MVP to 2 factories (**\$40K ARR**). Member of **Confluent AI Accelerator Cohort-3**.
- Built a **multi-agent scheduler** combining LLM agents with OR-Tools constraint solving, fed by **edge-deployed vision streaming** live counts and deviations, cutting weekly production replanning from 5 hours down to 30 minutes.
- Deployed a real-time multimodal assistant (LiveKit, voice + vision on iPad) with **RAG over machine manuals and SOPs**, enabling non-technical floor operators to replace 10K+ page manual lookups with hands-free voice queries.

Webbit Tech | *Website*

Oct 2022 – May 2025

Founder & Technical Lead

Macau

- Led technical delivery of 20+ web platforms, analytics dashboards, and AI pipelines to enterprise clients including **Hovione, Galaxy Entertainment Group, and Macau government agencies**.
- Shipped a ComfyUI-based **generative image pipeline** using SDXL and custom LoRA fine-tuning to produce game assets and e-commerce product visuals, reducing asset turnaround from **3 days to 0.5 days**.
- Managed client engagements end to end, including SLA authoring and compliance reviews against each client's own standards (pharma, gaming, government), scaling the team to 12 and **\$400K+ ARR** in 18 months.

Rainbow Seven Company Ltd.

Jun 2020 – Oct 2022

Data Scientist

Macau

- Cut reporting from every two months to daily by consolidating 30+ one-off SQL queries into a single **Python RPA extraction** over a 40-table custom backend, feeding Power BI dashboards leadership could filter and drill on demand.

Projects

Doctail.ai | *Website* | *Source Code*

Next.js | **PydanticAI** | **FastMCP** | **PostgreSQL**

- Built and deployed a **one-click MCP rollup platform** enabling developers to bundle multiple documentation sources into a single AI-queryable endpoint; onboarded 5 enterprise pilots and 20 beta users within 2 months of launch.
- Validated demand across 50+ developer interviews; wound down the product after Context7 consolidated the niche.

Propacks.net | *Website*

Next.js | **TypeScript** | **Redis** | **PostgreSQL** | **Medusa.js**

- Developed and deployed a headless B2B packaging marketplace with tiered pricing and automated fulfillment.
- Shipped a Hermes Agent SEO pipeline driving **3.9K MAU** and **5.8K ranked keywords** within **3 months**.

LoRA-based Efficient 3D Style Transfer | *Paper* | *Source Code*

PyTorch | **LoRA** | **CLIP** | **NeRF**

- Co-developed a parameter-efficient 3D style transfer method integrating LoRA adapters and CLIP embeddings into a modified LeRF/NeRF pipeline for efficient content-aware stylization.
- Achieved **10-25% higher PSNR** with 16x fewer parameters and 5x less GPU memory than StyleRF and FPRF. Project advised by **Hadar Averbuch-Elor** (Cornell) and **Yu-Lun Liu** (NYCU).

Education

Cornell University (Cornell Tech)

May 2025

Master of Engineering, Computer Science — GPA: 3.8/4.0

New York, NY

Newcastle University

May 2020

Bachelor of Science, Computer Science — GPA: 3.8/4.0

Newcastle, UK